



AURA

Academic Unified Response & Assistance

SOFTWARE ENGINEERING PROJECT

Milestone - 3 | Sep 2025 | Team 12

Our Team

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1. PROJECT SCHEDULE

1.1 Task Distribution

- The project's milestones are divided into smaller, manageable tasks. Each task is assigned to team members based on their expertise, availability, and workload to ensure a balance and efficient workflow.

Milestone	Tasks	Start Date	End Date	Assigned To
Milestone 1	Identify Users	01st Oct	05 Oct	All
	Define User Stories	04 Oct-25	14 Oct-25	All
	Draft and Submit Milestone 1 Report	15 Oct-25	20 Oct 2025	Nooha, Aryan, Taniya
Milestone 2	Design Storyboards	16 Oct 2025	18 Oct 2025	Nooha, Aryan, Mohit
	Design Low-Fidelity Wireframe	15 Oct 2025	19 Oct 2025	Taniya, Laxmi, Aryan, Aziz
	Draft and Submit Milestone 2 Report	17 Oct 2025	20 Oct 2025	Vikas, Taniya, Imran

1. PROJECT SCHEDULE

Milestone	Tasks	Start Date	End Date	Assigned To
Milestone 3	Project Scheduling using Gantt and Kanban Board	31 Oct 2025	02 Nov 2025	Mohit, Aziz
	Component Design	22 Oct 2025	27 Oct 2025	Vikas, Aryan
	Software Design - Class Diagram	30 Oct 2025	03 Nov 2025	Taniya,Noo ha
	Frontend Development	25 Oct 2025	30 Oct 2025	Vikas, Aryan, Taniya
	Draft and Submit Milestone 3 Report	01 Nov 2025	05 Nov 2025	All
Milestone 4	Design APIs	06 Nov 2025	09 Nov 2025	Laxmi, Mohit, Imran
	Create YAML File	10 Nov 2025	15 Nov 2025	Aryan, Vikas
	Integrate GenAI Features	08 Nov 2025	15 Nov 2025	Mohit, Imran, Aziz
	Draft and Submit Milestone 4 Report	13 Nov 2025	16 Nov 2025	All

1. PROJECT SCHEDULE

Milestone	Tasks	Start Date	End Date	Assigned To
Milestone 5	Prepare Test Cases for each API Endpoints	17 Nov 2025	20 Nov 2025	Aryan, Mohit,
	Test API Endpoints	21 Nov 2025	23 Nov 2025	Vikas, Imran
	Test GenAI Functionality	22 Nov 2025	27 Nov 2025	Aziz, Mohit
	Draft and Submit milestone 5 Report	25 Nov 2025	30 Nov 2025	Laxmi, Nooha, Taniya
Milestone 6	Complete Implementation and Final Report	01 Dec 2025	04 Dec 2025	Mohit, Vikas, Aziz
	Draft and Submit Final Report	04 Dec 2025	06 Dec 2025	Taniya, Imran, Aryan
	Prepare and Submit Final Presentation	06 Dec 2025	07 Dec 2025	Nooha, Laxmi, Mohit

1. PROJECT SCHEDULE

1.2 Sprint Backlog

SE Team 12

Search backlog

Filter

Understand user needs and document them using structured Agile tools.

SCRUM-16 Conduct user interviews

+ Epic

DONE

12 Oct

SCRUM-3 Create User Survey Form

+ Epic

DONE

14 Oct

SCRUM-17 Users identification

REQUIREMENT GATH...

DONE

5 Oct

SCRUM-4 Reach IIC Team for Requirements

REQUIREMENT GATH...

DONE

14 Oct

SCRUM-19 User Stories written using SMART guidelines.

REQUIREMENT GATH...

DONE

16 Oct

+ Create

SCRUM Sprint 2 - Milestone 2 10 Oct - 20 Oct (5 work items)

Design intuitive user interfaces and validate with feedback.

0 0 0 Complete sprint

SCRUM-5 Finalize the features

REQUIREMENT GATH...

DONE

15 Oct

SCRUM-6 Story board for the application - ppt or video format

UI/UX

DONE

18 Oct

SCRUM-7 low-fidelity wireframes with each user stories

+ Epic

DONE

18 Oct

SCRUM-9 Feedback from users regarding features and wireframe

+ Epic

DONE

19 Oct

SCRUM-10 Consolidate the wireframe is single pdf

+ Epic

DONE

20 Oct

+ Create

SCRUM Sprint 3 - Milestone 3 21 Oct - 5 Nov (8 work items)

Finalize architecture, scheduling, and frontend development.

Complete sprint

SCRUM-11 Design of Components

+ Epic

DONE

30 Nov

SCRUM-13 Software designing - class diagram

+ Epic

DONE

3 Nov

SCRUM-41 Student (AI assistance dashboard UI)

UI/UX

DONE

30 Oct

SCRUM-42 Student(My course and resource dashboard UI)

UI/UX

DONE

30 Oct

SCRUM-43 Student(My Query and dashboard UI)

UI/UX

DONE

30 Oct

SCRUM-16 Gantt chart and kanban board

SCRUM

DONE

2 Nov

SCRUM-12 Sprint Schedule

SCRUM

IN PROGRESS

SCRUM-14 Details of MoM

SCRUM

IN PROGRESS

+ Create

SCRUM Sprint 4 - Milestone 4 6 Nov - 16 Nov (10 work items)

Implement backend APIs and perform functional testing.

Start sprint

SCRUM-63 Setup Backend

DEVELOPMENT

10 000+

6 Nov

SCRUM-64 Design API Endpoints

DEVELOPMENT

10 000+

6 Nov

SCRUM-82

DEVELOPMENT

10 000+

11 Nov

SCRUM-83

DEVELOPMENT

10 000+

12 Nov

SCRUM-65 Add Error Handling and Logging

+ Epic

10 000+

10 Nov

SCRUM-66 Write Unit Tests

DEVELOPMENT

10 000+

9 Nov

SCRUM-67 Document Test Results

DEVELOPMENT

10 000+

11 Nov

SCRUM-68 Create Swagger-Compatible YAML File

DEVELOPMENT

10 000+

12 Nov

SCRUM-69 Prepare Test Case Document (PDF)

DEVELOPMENT

10 000+

13 Nov

SCRUM-70 Compile Milestone 4 Document

DEVELOPMENT

10 000+

14 Nov

+ Create

SCRUM Sprint 4 - Milestone 4 6 Nov - 16 Nov (10 work items)

Implement backend APIs and perform functional testing.

Start sprint

SCRUM-63 Setup Backend

DEVELOPMENT

10 000+

6 Nov

SCRUM-64 Design API Endpoints

DEVELOPMENT

10 000+

6 Nov

SCRUM-82

DEVELOPMENT

10 000+

11 Nov

SCRUM-83

DEVELOPMENT

10 000+

12 Nov

SCRUM-65 Add Error Handling and Logging

+ Epic

10 000+

10 Nov

SCRUM-66 Write Unit Tests

DEVELOPMENT

10 000+

9 Nov

SCRUM-67 Document Test Results

DEVELOPMENT

10 000+

11 Nov

SCRUM-68 Create Swagger-Compatible YAML File

DEVELOPMENT

10 000+

12 Nov

SCRUM-69 Prepare Test Case Document (PDF)

DEVELOPMENT

10 000+

13 Nov

SCRUM-70 Compile Milestone 4 Document

DEVELOPMENT

10 000+

14 Nov

+ Create

SCRUM Sprint 5 - Milestone 5 17 Nov - 30 Nov (10 work items)

Prepare a detailed report including milestones, implementation details, tools used, and user instruction.

Start sprint

SCRUM-71 Prepare API Test Cases Document

DEVELOPMENT

10 000+

16 Nov

SCRUM-72 Capture Test Execution Screenshots

DEVELOPMENT

10 000+

16 Nov

SCRUM-73 Complete Final PDF Report

+ Epic

10 000+

16 Nov

+ Create

SCRUM Sprint 6 - Milestone 6 30 Nov - 7 Dec (8 work items)

Final delivery

Start sprint

SCRUM-74 Complete Project Implementation & Post-mortem

DEVELOPMENT

10 000+

25 Dec

SCRUM-75 Complete a detailed report including milestones, implementation details, tools used, and user instruction

DEVELOPMENT

10 000+

22 Dec

SCRUM-76 Document Implementation Details

+ Epic

10 000+

22 Dec

SCRUM-77 Add screenshots and descriptions of code review process, issue reporting, and tracking mechanisms

DEVELOPMENT

10 000+

22 Dec

SCRUM-78 Prepare Presentation Slides

DEVELOPMENT

10 000+

24 Dec

SCRUM-79 Record Project Demonstration Video

+ Epic

10 000+

24 Dec

SCRUM-80 Package Complete Codebase

+ Epic

10 000+

24 Dec

SCRUM-81 Complete Final PDF Report

DEVELOPMENT

10 000+

24 Dec

+ Create

SCRUM Sprint 7 - Milestone 7 30 Nov - 7 Dec (8 work items)

Final delivery

Start sprint

SCRUM-82 Design API

DEVELOPMENT

10 000+

25 Dec

SCRUM-83 Prepare Milestone 7 Report

DEVELOPMENT

10 000+

25 Dec

SCRUM-84 Create Project Schedule

DEVELOPMENT

10 000+

25 Dec

SCRUM-85 Build UML Class Diagram

DEVELOPMENT

10 000+

25 Dec

SCRUM-86 Prepare Milestone 7 Report

DEVELOPMENT

10 000+

25 Dec

SCRUM-87 Create Gantt chart and Kanban Board

DEVELOPMENT

10 000+

25 Dec

SCRUM-88 Prepare unit tests for each API endpoint

DEVELOPMENT

10 000+

25 Dec

SCRUM-89 Test API endpoints

DEVELOPMENT

10 000+

25 Dec

SCRUM-90 Record Setup

DEVELOPMENT

10 000+

25 Dec

SCRUM-91 Deployment of application

DEVELOPMENT

10 000+

25 Dec

SCRUM-92 Milestone 7 report creation

DEVELOPMENT

10 000+

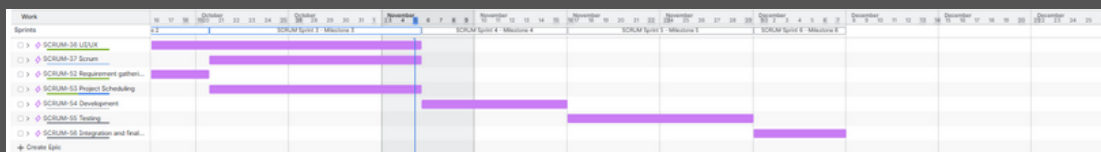
25 Dec

+ Create

1. PROJECT SCHEDULE

1.3 Gantt Chart

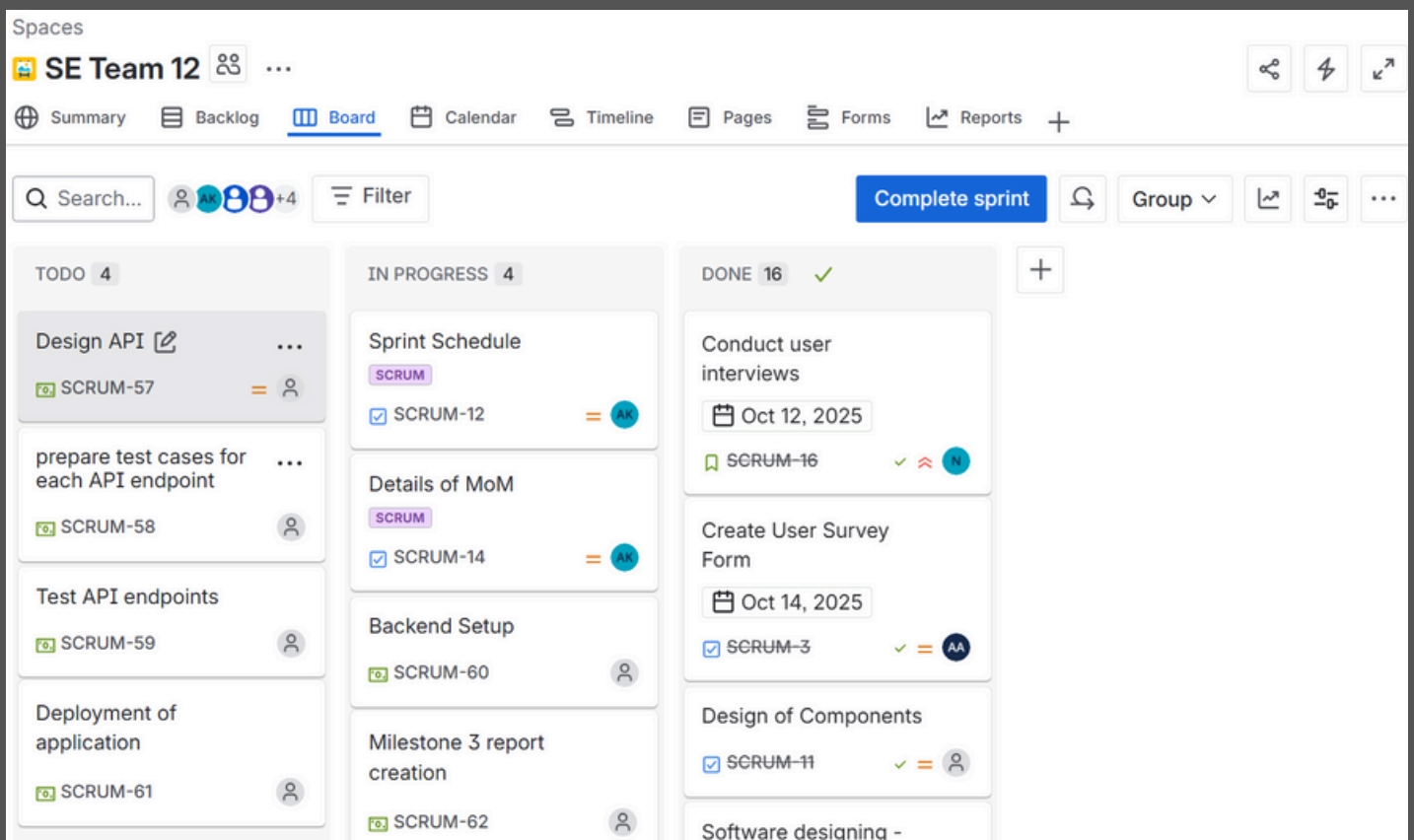
- A Gantt Chart was created in Jira to represent the project's timeline and progress, divided into multiple sprints. This helped in visually organizing tasks, their durations, dependencies, and completion status, ensuring that all team members are aligned with the milestones and deadlines.



1. PROJECT SCHEDULE

1.4 Kanban Board

- The team utilized a Kanban board in Jira to efficiently manage sprint tasks, monitor progress, and ensure clear accountability throughout Milestone 3. Each task was categorized under columns such as To Do, In Progress, Review, and Done, reflecting the real-time development status across all modules – Student, TA, Instructor, and Admin.
- The board helped maintain continuous visibility of the project workflow, enabling the team to prioritize issues, resolve blockers promptly, and achieve balanced workload distribution.
- By the end of this milestone, the Kanban board reflected substantial progress, with key tasks for UI implementation, component design, and documentation successfully transitioned to the Done phase, ensuring timely milestone completion.



(SCRUM-1 to SCRUM-3) — Team Formation & Orientation (17 Sep – 23 Sep)

Focus: Team introduction, role assignment, and domain exploration.

Summary:

- Team members introduced themselves and discussed prior experience with Vue.js, Python, and database technologies.
- Reviewed previous year project reports and problem statements for inspiration.
- Discussed project domains and finalized Customer Service as the broad sector for exploration.
- Assigned preliminary roles based on technical strengths (Frontend, Backend, Documentation, UI/UX).
- **Action Items:**
 - Each member to research ideas from different sectors (education, healthcare, e-commerce, academic services).
 - Prepare short presentations for the next scrum to pitch potential project ideas.

2. SCRUM MEETINGS

(SCRUM-4 to SCRUM-6)— Problem Statement & Idea Finalization (02 Oct – 10 Oct)

Focus: Idea evaluation, user identification, and initial user story collection.

Summary:

- Team explored all presented ideas and narrowed focus to the Academic Service domain under customer service.
-
- Defined the problem statement around “AI-powered academic service automation.”
- Conducted user interviews with students, TAs, and instructors to collect initial user stories.
- Set up Jira Kanban Board for task tracking and discussed agile workflow.
- Drafted initial Milestone 1 submission report and presented key ideas to the client.
- The client emphasized focusing on provider-side challenges and improving user story clarity.

Action Items:

1. Finalize SMART-based problem statement.
2. Conduct additional interviews focusing on secondary users (TAs, instructors, and admins).
3. Refine user stories and categorize by user type (Primary, Secondary, Tertiary).

2. SCRUM MEETINGS

2.2 User Interfaces

(SCRUM-7 to SCRUM-9)— Feature Finalization & Storyboarding (10 Oct – 20 Oct)

Focus: Refining user stories, defining features, and preparing for wireframe creation.

Summary:

- Collected more user stories from secondary users (TAs and Instructors).
- Finalized ten AI-based features based on story alignment and feasibility.
- Client suggested adding two additional features (Query Escalation System and Analytics Dashboard).
- Began drafting wireframes and storyboards for selected features based on finalized user stories.
- Started organizing user stories under relevant features and GenAI roles.

Action Items:

1. Submit the updated feature list and user story document for client review.
2. Begin visual design for storyboard representation.
3. Plan project milestones and timelines for Milestone 2.

2. SCRUM MEETINGS

Sprint 3 — UI/UX Implementation and Component Design (21 Oct – 04 Nov)

Focus: Project planning, frontend implementation, and component design for Milestone 3 deliverables.

Summary:

- The team created a comprehensive Project Plan outlining sprint structure, milestone deliverables, and timelines to maintain clarity and accountability across roles.
- Defined a consistent UI theme and design style, following IITM's institutional color palette and academic branding standards.
- Developed and presented the Class Diagram as part of Milestone 3, ensuring system structure alignment with finalized features.
- Implemented frontend modules for Student, TA, and Instructor dashboards — achieving approximately 70% completion with routing, left-side navigation, and responsive design.
- Began component diagram development and frontend integration work, using dummy data for interface testing.
- Prepared internal Scrum Minutes and GitHub issue logs to maintain transparent communication and task tracking.

Action Items:

1. Submit Project Plan to the client for feedback and milestone verification.
2. Review and refine the Class Diagram by 31 October 2025 to incorporate client feedback.
3. Design and finalize 7 key component diagrams (Dashboard, Chat Interface, Query Tracker, Analytics Panel, Slide Generator, Content Recommender, and Onboarding Assistant).
4. Conduct internal UI testing and review, ensuring consistent navigation, responsiveness, and error-state handling.
5. Prepare and submit Milestone 3 Report by 4 November 2025, including screenshots, diagrams, and the frontend code package (with README instructions).

3. SOFTWARE DESIGN - SYSTEM COMPONENTS

Base URL Convention

All routes prefixed as:

/api/v1/

1.Authentication & User Management

Method Endpoint Description

POST /auth/register Register new user (student/TA/instructor/admin)

POST /auth/login Login and return JWT access token

POST /auth/logout Invalidate token / logout

GET /auth/me Get current user profile

PATCH /auth/update Update user profile details

POST /auth/refresh Refresh access token

2.AI Chat & Query Handling (RAG Assistant)

Method Endpoint Description

POST /chat/ask Ask academic/admin question (RAG-powered chatbot)

GET /chat/history Retrieve user's previous chat sessions

POST /chat/feedback Submit feedback for AI response

GET /chat/context Retrieve context docs or references used in last answer

API endpoints 1

3.Smart Study Assistant(Summaries,Deadlines,Reminders)

Method Endpoint Description

GET /study/summary/weekly

Generate or fetch weekly summary of lectures, quizzes, announcements

POST /study/reminders/create Create smart study-break reminder

PATCH /study/reminders/update Update smart study-break reminder

DELETE /study/reminders/delete Delete smart study-break reminder

GET /study/reminders List all active reminders

POST /study/deadlines/add Add project, quiz, or admin deadline

GET /study/deadlines Get user's academic calendar

3. SOFTWARE DESIGN - SYSTEM COMPONENTS

4. AI Content Recommender

Method Endpoint Description

GET /recommend/content

Fetch AI-based learning resource recommendations

GET /recommend/related Suggest external resources for given topic

POST /recommend/feedback Log user engagement or usefulness feedback

5. TA Doubt Summarizer

Method Endpoint Description

POST /ta/doubts/upload Upload student chat/forum/email data

GET /ta/doubts/summary Get summarized and clustered student doubts

GET /ta/doubts/topics View most confusing topics (cluster analysis)

GET /ta/doubts/insights Get LLM-driven insights on learning gaps

6. AI Query Escalation & Tracking

API endpoints 2

Method Endpoint Description

POST /queries/submit Submit new academic/admin/technical query

GET /queries View user queries and status

GET /queries/unresolved For TA/Admin: fetch unresolved queries

PATCH /queries/assign/:id Assign query to TA/Instructor

PATCH /queries/status/:id Update query status (pending/resolved/escalated)

7. Admin Analytics Dashboard

Method Endpoint Description

GET /analytics/overview Get overall metrics (queries, responses, users)

GET /analytics/faqs Top FAQs or recurring questions

GET /analytics/performance Response time and resolution rate

GET /analytics/sentiment Aggregate feedback sentiment

GET /analytics/usage Active users, API calls, session stats

3. SOFTWARE DESIGN - SYSTEM COMPONENTS

8.AI Discourse Integration (Forums & Threads)

Method Endpoint Description

GET /discourse/threads List forum discussions by topic

GET /discourse/summary/:topic Summarize discussion threads on a topic

GET /discourse/search Semantic search through forum history

POST /discourse/link Link related discussions together

9.TA Onboarding Mentor

Method Endpoint Description

POST /mentor/chat Ask onboarding or teaching-related questions

GET /mentor/resources Get curated onboarding materials

GET /mentor/tips Fetch daily or contextual teaching tips

API endpoints 3

10.Automated Assessment Generator

Method Endpoint Description

POST /assessment/generate Generate question paper by topic/difficulty

GET /assessment/templates Fetch available question templates

PATCH /assessment/edit/:id Modify generated questions manually

POST /assessment/export Export final question paper (PDF/DOCX)

11.Slide Deck Generator

Method Endpoint Description

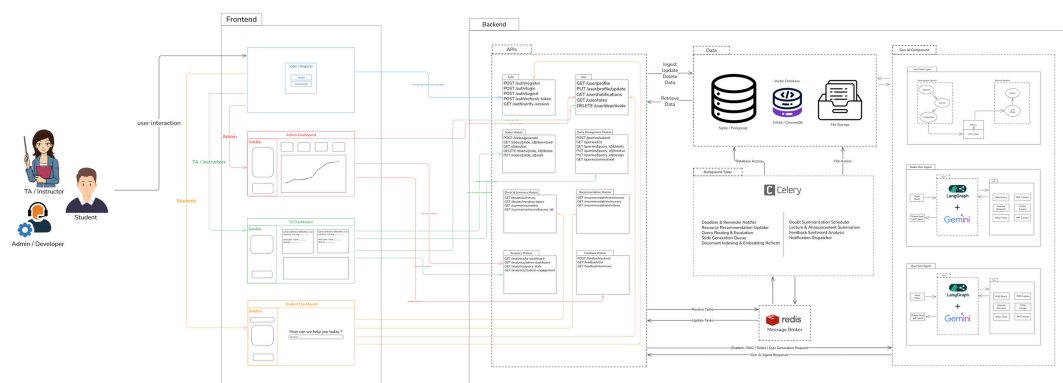
POST /slides/generate Generate teaching slides from structured content

GET /slides/history View generated slide deck history

POST /slides/upload Upload custom templates

GET /slides/download/:id Download generated slide deck

Architecture Design



3. SOFTWARE DESIGN - SYSTEM COMPONENTS

3.1 Frontent

TA/Instructor View (Query Response & Content Creation)

- View assigned queries
 - Dashboard of queries assigned to them
 - Filter by priority, category, and status
 - View student query history
- Respond to queries
 - Start chat sessions for query resolution
 - Update query status (In Progress, Resolved)
 - Add detailed responses with file attachments
- Generate educational content
 - Use AI to create slide decks from topics
 - Generate assessments with customizable questions
 - Upload and manage learning resources
- Monitor student engagement
 - View query response times
 - Track student participation in chats
 - Access content usage analytics
- Manage notifications
 - Set preferences for query alerts
 - Receive updates on new queries and deadlines

Student View (Query Management & Learning)

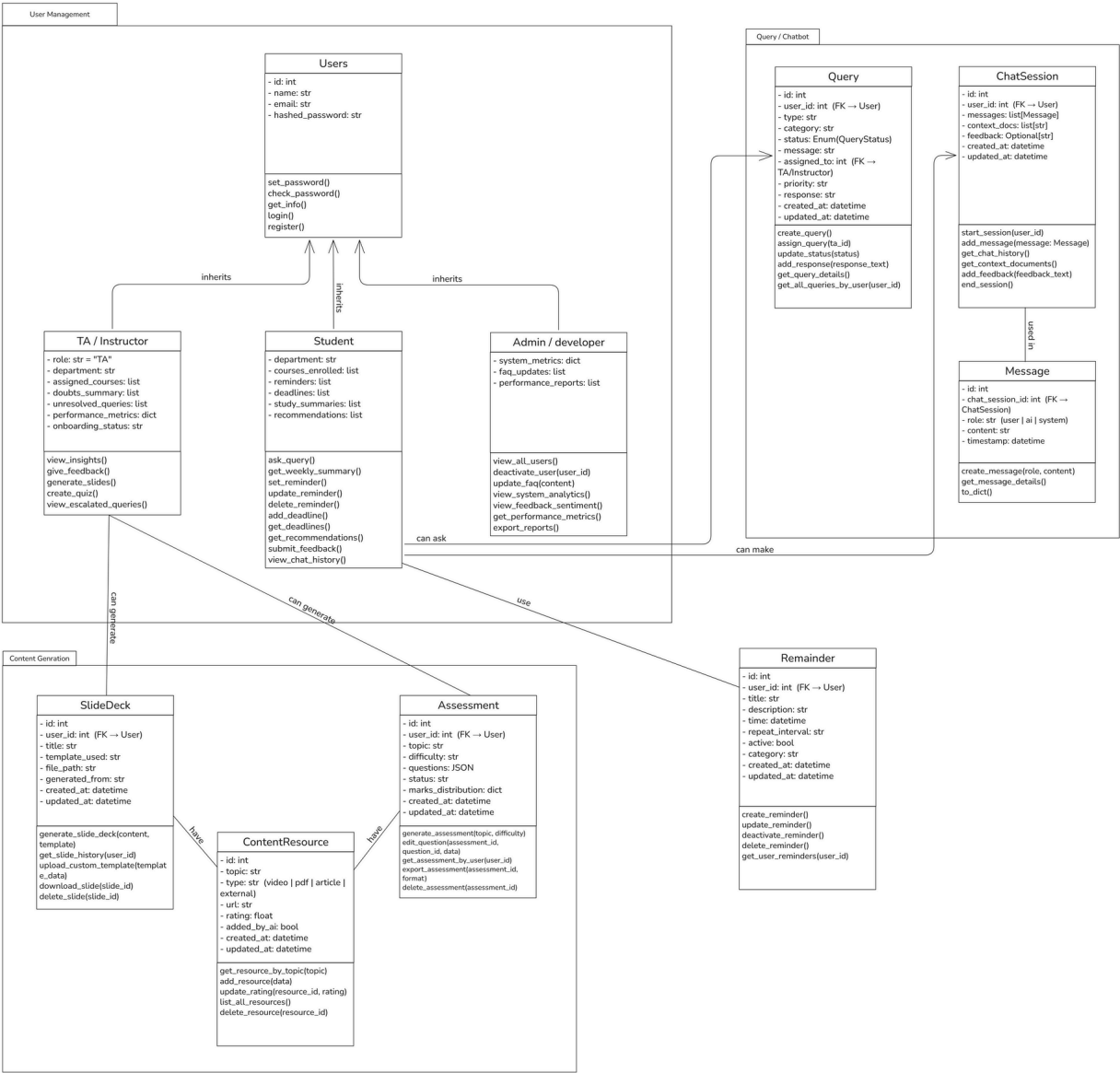
- Create a new query
 - Select category (academic, technical, general)
 - Set priority level
 - Add message description with file attachments
- View and manage queries
 - Dashboard to view all submitted queries
 - Filter by status (Assigned, In Progress, Resolved)
 - Track query assignment to TA/Instructor
- Chat with TA/Instructor
 - Real-time messaging for query resolution
 - View chat history and context documents
 - Receive notifications for responses
- Access learning resources
 - Browse slide decks, assessments, and materials
 - Download content for offline access
 - Rate and provide feedback on resources
- Manage reminders
 - Create reminders for deadlines and exams
 - Receive notifications for query updates
 - View reminder dashboard

Admin/Developer View (System Management)

- User management
 - Create and manage user accounts
 - Assign roles (Student, TA, Instructor, Admin)
 - View user activity logs
- System monitoring
 - View platform usage statistics
 - Monitor query resolution metrics
 - Track AI content generation usage
- Content moderation
 - Review and approve user-generated content
 - Manage content categories and tags
 - Handle reported issues
- System configuration
 - Update notification templates
 - Configure AI service parameters
 - Manage database backups and storage
- Analytics and reporting
 - Generate system-wide performance reports
 - Export data in PDF/CSV formats
 - View user engagement trends

Class Diagram

Class Diagram



UI Pages

AURA

Dashboard

Query Tracker

Doubt Summarizer

Resource Hub

Onboarding Mentor

Assessment Generator

Slide Deck Creator

Profile

Search courses, threads, or resources

Ask AI

TA

Welcome back, TA! 🌟

Here's your teaching activity overview

Doubts Resolved

24

This week

Students Helped

18

This week

Avg Response Time

45 mins

Improved by 20%

Satisfaction Rate

92%

From student feedback

Assigned Courses

Software Engineering

85C3301

45

Students

Machine Learning

85C3303

38

Students

Recent Student Doubts

Help with Milestone 3 design patterns

new doubt • 85C3301 • 30 mins ago

pending

Report

Neural network convergence issue

new Doubt • 85C3303 • 2 hours ago

answered

Report

Database normalization clarification

new doubt • 85C3301 • 45 mins ago

pending

Report

Escalated Queries

Grade re-evaluation request

new student's Escalated to instructor

2024-10-30

Extension request for medical reasons

Sarah Lee • Escalated to Admin

2024-10-29

Quick Actions

Add New Resource

Summarize Doubts

View All Queries

Start Onboarding Session

Recent Announcements

New Feature: AI Summarizer

The new doubt summation is now live! Explore its capabilities.

System Maintenance

Scheduled maintenance on 15th May, 10 PM - 12 AM IST

Resource HUB Updates

Latest academic papers for Advanced AI course added.

AURA

Dashboard

Query Tracker

Doubt Summarizer

Resource Hub

Onboarding Mentor

Assessment Generator

Slide Deck Creator

Profile

Search courses, threads, or resources

Ask AI

TA

Export to PDF

Share Link

Current Slide: Introduction to EduSlide Gen

Prev

Next

Type your prompt here to generate or refine slides...

Generate Slides

Welcome slide!

Reorder Slides (Drag & Drop)

1

Introduction to EduSlide Gen

Welcome slide!

2

Agenda

Topics covered ...

3

Core Concepts

Explain concepts ...

4

Example Problems

Practice slides ...

AURA

Dashboard

Query Tracker

Doubt Summarizer

Resource Hub

Onboarding Mentor

Assessment Generator

Slide Deck Creator

Profile

Search courses, threads, or resources

Ask AI

TA

Resource Hub

All

Mathematics

Programming

Soft Skills

AI/ML

Academic Writing

Ethics

Productivity

Academic Tools

Search resources...

Add New Resource

Request Resource

Mathematics

Introduction to Calculus

A comprehensive guide covering limits, derivatives, and integrals.

View Details

Programming

Advanced Python for Data Science

Dive deep into Python's data science ecosystem. Covers NumPy, pandas, and more.

View Details

Soft Skills

Effective Presentation Skills

Learn how to deliver impactful academic presentations. Includes examples.

View Details

Academic Writing

Research Paper Writing Workshop

A step-by-step workshop on structuring, writing, and formatting research papers.

View Details

Productivity

Online Collaboration

Explore various digital tools to enhance teamwork and communication.

View Details

Ethics

Ethics in academics

A crucial guide for its on upholding and promoting academic integrity.

View Details

AI/ML

Foundations of Machine Learning

An introductory video series on machine learning concepts.

View Details

Academic Tools

LaTeX for Academics

Master LaTeX for typesetting high-quality academic documents and research reports.

View Details

Sign Out

logged in as TA

AI Onboarding Mentor

Your personalized guide trained on institutional best practices, senior TA experiences, and faculty wisdom

AI Mentor Chat

Connect with TA/faculty

Quick Start Guide

Senior TA Tips

Getting Started as a TA

Learn the fundamentals of your TA responsibilities

Common Questions

- What are my main responsibilities?
- How do I communicate with students effectively?
- What tools do I need to use?

Handling Student Queries

Best practices for responding to doubts

Common Questions

- How to prioritize urgent queries?
- Best response time expectations?
- When should I escalate to instructor?

Time Management

Balance TA work with your coursework

Common Questions

- How many hours should I dedicate weekly?
- Tips for managing multiple courses?
- Avoiding burnout strategies?

Grading Best Practices

Fair and consistent assessment guidelines

Common Questions

- How to grade assignments fairly?
- Handling re-evaluation requests?
- Providing constructive feedback?

Common Challenges

Solutions to typical TA difficulties

Common Questions

- Dealing with difficult students?
- Managing large query volumes?
- Handling academic integrity issues?

Office Hours Tips

Making office hours effective

Common Questions

- How to structure office hours?
- Handling multiple students at once?
- Virtual vs in-person office hours?

AURA

- Dashboard
- AI Assistant
- My Courses & Resources
- My Queries
- Profile

QUICK ACTIONS

- New Query
- Video Summary

Sign Out

Welcome back, Aryan!

Here's what's happening with your academics today

12

3

2

18

Upcoming Deadlines

Software Engineering Project 1 Milestone 3
BCS3081
Due: 2024-11-05

Pending

Data Science Assignment - Module 2
BCS2015
Due: 2024-11-07

Pending

AI Quiz - Unit 4
BCS3085
Due: 2024-11-10

Pending

Recent Accomplishments

Mid-term exam schedule updated
BCS3081
2 days ago

New project guidelines released
BCS2015
1 day ago

Assignment 2 evaluation complete
BCS3085
3 days ago

Course Progress

Software Engineering
BCS3081

75%

Data Science Fundamentals
BCS2015

60%

Artificial Intelligence
BCS3085

40%

AURA

- Dashboard
- AI Assistant
- My Courses & Resources
- My Queries
- Profile

QUICK ACTIONS

- New Query
- Video Summary

Sign Out

Logged in as Student

Search courses, threads, or resources

Ask AI



Welcome!

Ask about deadlines, resources, or concepts. I can summarize lectures and draft study plans.

What are my upcoming deadlines this week?



Upcoming deadlines

CS 301 • Lab 2 writeup

Today

CS 210 • Project 2 milestone

Mon 11:59 PM

Suggested Prompts

Summarize last CS 301 lecture

Explain Dijkstra with example

Draft study plan for finals

Find office hours this week

These are tailored to your enrolled courses.

Context

Type a follow-up...

Send

AURA

Dashboard

AI Assistant

My Courses & Resources

My Queries

Profile

QUICK ACTIONS

New Query

Video Summary

Sign Out

Logged in as Student

Search courses, threads, or resources

Ask AI

TA

4 Active Courses | 3 Deadlines This Week

AllEnrolledArchivedAdd ResourcesStudy Break

Data Structure and Algorithms

CS 301: Algorithms

Mon/Wed • 10:00–11:15

Prof. Nguyen

OpenAsk AIEnrolled

CS 210: Systems

Tue/Thu • 9:00–10:15

Prof. Clarke

OpenAsk AIArchived

A visualization of matrix manipulation

MATH 241: Linear Algebra

Tue/Thu • 1:00–2:15

Dr. Patel

OpenAsk AIEnrolled

CS301 • Lab 2 rubric

Systems • Lecture 5

Linear Algebra • Notes

Quick actions

View deadlines

Share to class

Pinned resources

Continue where you left

CS301 Thread • Graphs

Systems Lab • Setup

AURA

Dashboard

AI Assistant

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QUICK ACTIONS

New Query

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Search courses, threads, or resources

Ask AI

TA

Search My QueriesAllOpenResolvedNew Query

Recent queries7 this month

Help with Dijkstra complexity

CS 301 • 2h ago

I implemented the algorithm but still see $O(n^3)$...

Open

Systems lab Docker error

CS 210 • Yesterday

Docker build fails on ARM machine...

Resolved

Office hour booking confirmation

HOST • 1d ago

Want to confirm the location for Friday...

Open

CS 301AlgorithmsShareMark Resolved

Dijkstra complexity on dense graphs

Created 2h ago • Query ID: Q-87134

YouOpen

I used adjacency matrix + binary heap and got $O(n^3)$. Is this expected for dense graphs?

45 min ago

AI Assistant

Yes. With an adjacency matrix the edge scans dominate at $O(n^3)$... Use adjacency lists; on dense graphs, a Fibonacci heap won't help asymptotically versus array/heap implementations.

45 min ago

TA Lee

Good afternoon! I'm glad you're working on this. For dense graphs, the standard adjacency matrix approach is indeed $O(n^3)$. While a Fibonacci heap can improve the runtime to $O(n^2 \log n)$ for sparse graphs, it doesn't help with the asymptotic complexity for dense graphs. Using adjacency lists and a binary heap, you can achieve $O(n^2 \log n)$, which is a significant improvement over the naive $O(n^3)$ approach.

Filters

CourseAll

StatusOpen

DateThis month

Suggested articles

Dijkstra variants

Dense vs sparse graphs

Type a follow-up...

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Video Summary Generator

Enter Video Link (YouTube, Vimeo, etc.)

https://www.youtube.com/watch?v=...

Generate

Summary

Enter a video link above to get the summary.

Welcome back, Admin! 🙌

Here's a birds-eye view of your institution for today.

1,276

Active Users

86

Open Queries

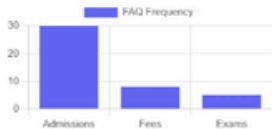
19

Flagged Issues

99%

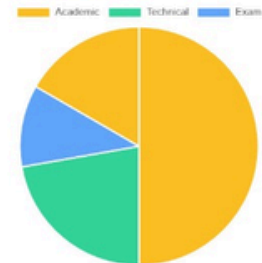
System Health

FAQ Analytics



Top-asked questions and answer accuracy.

Unresolved Queries



Monitor backlog and escalation rate.

Quick Actions

Send Announcement

Export Reports

Manage Roles

Recent System Announcements

- Resource HUB update added for 'Advanced AI'
- New feature: AI Query Summarizer is now live!
- Scheduled Maintenance: Nov 10, 12 AM – 2 AM

Recent Institutional Queries

Export

ID	Topic	Role	Date	Status
3012	Exam Dates Change	Student	2025-11-03	Pending
1989	Resource Access Issue	Instructor	2025-11-02	Resolved
2244	Duplicate Queries	TA	2025-11-01	Escalated
2967	Policy Suggestion	Student	2025-11-01	Pending

User Management

Bulk Approve

User	Email	Role	Status	Actions
Jane Doe	jane@iitm.ac.in	TA	Pending	Approve Block Flag
Dr. S. Kalra	kalra@iitm.ac.in	Instructor	Active	Block Flag
Ankur Sharma	ankur@iitm.ac.in	Student	Blocked	Flag
Priya Mehta	priya@iitm.ac.in	Student	Active	Block Flag

© 2025 Institute Admin Panel

AURA

- Dashboard
- Discussion Summaries
- Class Schedule
- Question Paper Generator
- Profile

Welcome back, Instructor! 🙌

Here's an overview of what's happening in your classroom today.

Student Engagement Over Time

View Details



Assignment Performance Heatmap

Student	View All Assignments				
	A1	A2	A3	A4	A5
Alice Smith	85	78	92	88	80
Bob Johnson	75	82	85	79	83
Carol White	90	88	80	85	87
David Green	88	85	82	86	84
Eve Black	82	80	85	83	81

Upcoming & Missed Deadlines

View All Deadlines

Project Proposal	Due: 10/25/2024	Overdue
Mid-term Exam	Nov 10, 2024	Upcoming
Research Paper Draft	Nov 15, 2024	Upcoming
Final Project Submission	Dec 01, 2024	Upcoming
Course Feedback	Dec 15, 2024	Upcoming

Quiz Score Distribution

View All Scores



Quick Actions

- Generate Progress Report
- Export Student Data
- View Student Overview
- Create New Assignment

Top Performing Students

Student	Score (95)
Alice Smith	Score 95
Alexandra Singh	Score 95
Olivera Patel	Score 95
Priya Mehta	Score 95
Goldsmith-Park	Score 95

Discussion Summaries

Overview of student discussions and feedback sentiment

Total Discussions

100

Active Threads

15

Average Sentiment

75%

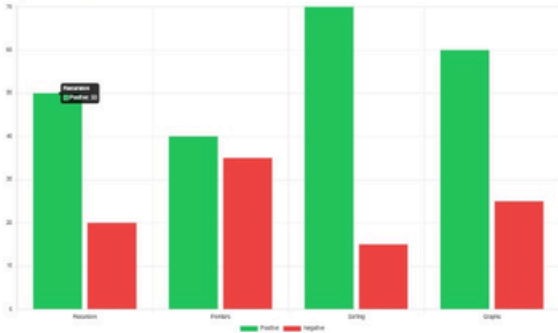
Unresolved Queries

5

Sentiment Trend Over Time



Key Sentiment Drivers



Top Discussion Topics

Understanding Recursion

Students discuss the conceptual depth of recursive algorithms.

Participants: Alice, Bob, Charlie

60% Positive

20% Neutral

10% Negative

[View Full Thread](#)

Data Structures - Linked List

Common confusion around memory allocation and pointers.

Participants: David, Emily, Frank

55% Positive

30% Neutral

15% Negative

[View Full Thread](#)

Sentiment Insights

Positive Insights

- Students found the recursion examples engaging.
- Clear visualization improved understanding.

[View Full Thread](#)

Neutral Insights

- Some students suggested slower pacing.
- Clarified syntax but still unsure of logic depth.

[View Full Thread](#)

Negative Insights

- Students struggled with base case logic.
- Need more practice problems for reinforcement.

[View Full Thread](#)